Hartree-Fock-Nambu mean-field theory

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A. Introduction

Consider an interacting many-body system of electrons described by the Hamiltonian

$$H = H_0 + H_{\text{int}} = \sum_{\alpha\beta} h_{\alpha\beta} c^\dagger_\alpha c_\beta + \sum_{\alpha\beta\gamma\delta} V^{\alpha\beta}_{\gamma\delta} c^\dagger_\alpha c^\dagger_\beta c_\gamma c_\delta, \tag{1}$$

with $c^\dagger(c)$ fermionic creation (annihilation) operators and the Greek indices α, β, γ, δ generic degrees of freedom. The time-dependency on the fermionic operators are being omitted for clarity, as is also done in the non-interacting Hamiltonian, accounting for a coupling to a perturbative external force, i.e $h_{\alpha\beta} \to h_{\alpha\beta} + B_{\alpha\beta}F(t)$. Concerning the interaction term, we assume $V(\mathbf{r})$ to be generic electrostatic potential of symmetry $V(\mathbf{r}-\mathbf{r}') = V(\mathbf{r}'-\mathbf{r})$ such that the interacting potential tensor has the symmetries $V^{\alpha\beta}_{\gamma\delta} = V^{\beta\alpha}_{\delta\gamma}$ and $(V^{\alpha\beta}_{\gamma\delta})^* = V^{\gamma\delta}_{\alpha\beta}$.

Symmetry properties of V
From the imposed symmetry $V(\mathbf{r}-\mathbf{r}') = V(\mathbf{r}'-\mathbf{r})$, we obtain

$$\begin{aligned} V^{\alpha\beta}_{\gamma\delta} &= \int d\mathbf{r}d\mathbf{r}' \phi^*_\alpha(\mathbf{r})\phi^*_\beta(\mathbf{r}')V(\mathbf{r}-\mathbf{r}')\phi_\gamma(\mathbf{r}')\phi_\delta(\mathbf{r}),\ \mathbf{r} \leftrightharpoons \mathbf{r}' \\ &= \int d\mathbf{r}d\mathbf{r}' \phi^*_\alpha(\mathbf{r}')\phi^*_\beta(\mathbf{r})V(\mathbf{r}-\mathbf{r}')\phi_\gamma(\mathbf{r})\phi_\delta(\mathbf{r}') \\ &= \int d\mathbf{r}d\mathbf{r}' \phi^*_\beta(\mathbf{r})\phi^*_\alpha(\mathbf{r}')\, V(\mathbf{r}-\mathbf{r}')\phi_\delta(\mathbf{r}')\phi_\gamma(\mathbf{r}) = V^{\beta\alpha}_{\delta\gamma} \ \square \end{aligned}$$

Similarly, we can show that

$$\begin{aligned} \left(V^{\alpha\beta}_{\gamma\delta}\right)^* &= \int d\mathbf{r}d\mathbf{r}' \phi_\alpha(\mathbf{r})\phi_\beta(\mathbf{r}')V(\mathbf{r}-\mathbf{r}')\phi^*_\gamma(\mathbf{r}')\phi^*_\delta(\mathbf{r}) \\ &= \int d\mathbf{r}d\mathbf{r}' \phi^*_\delta(\mathbf{r})\phi^*_\gamma(\mathbf{r}')V(\mathbf{r}-\mathbf{r}')\phi_\delta(\mathbf{r}')\phi_\alpha(\mathbf{r}) = V^{\delta\gamma}_{\beta\alpha} = V^{\gamma\delta}_{\alpha\beta} \ \square \end{aligned}$$

In order to study the dynamics of this system, we solve for the time-evolution of the the system's reduced density matrix (rDM) through it's its equation of motion (EoM). The system's purely electronic reduced density matrix (rDM) is defined as $\rho^{\text{ee}}_{ba} = \langle c^\dagger_a c_b \rangle$. However, if one wishes to study superconducting systems, one must also account for the anomalous terms. For this, we introduce the Nambu spinor representation with doubling of degrees of freedom $\check{c}_a = (c_a\ c^\dagger_a)$. The Nambu representation of the rDM, can then be written as the tensor product of this spinors for each of the spin combinations,

$$\rho_{ab} = \begin{pmatrix} \langle c^\dagger_b c_a \rangle & \langle c_b c_a \rangle \\ \langle c^\dagger_b c^\dagger_a \rangle & \langle c_b c^\dagger_a \rangle \end{pmatrix} = \begin{pmatrix} \rho^{\text{ee}}_{ab} & \rho^{\text{eh}}_{ab} \\ \rho^{\text{he}}_{ab} & \rho^{\text{hh}}_{ab} \end{pmatrix}. \tag{2}$$

Directly from the (equal-time) fermionic anti-commutator properties, this terms relate to each other as $\rho^{\text{hh}}_{ab} = -\rho^{\text{ee}}_{ba} + \delta_{ab}$ and $\rho^{\text{he}}_{ab} = (\rho^{\text{eh}}_{ba})^*$, but also $\rho^{\text{ee/hh}}_{ab} = (\rho^{\text{ee/hh}}_{ba})^*$ and $\rho^{\text{eh/he}}_{ba} = -\rho^{\text{eh/he}}_{ab}$. Notice that, since the spin degree of freedom was not yet specified we are not yet restricting ourselves to typical s-wave spin-singlet superconductors, this is a general treatment. Our task now is then to solve the two distinct EoMs, for the purely electronic rDM ρ^{ee}_{ab}, which then determines the other two simple through their relations.

B. Reduced density matrices equations of motion

We start by solving for the purely electronic rDM equation of motions

$$\frac{d}{dt}\rho^{\text{ee}}_{ab}(t) = \left\langle \frac{dc^\dagger_b(t)}{dt} c_a(t) \right\rangle + \left\langle c^\dagger_b(t) \frac{dc_a(t)}{dt} \right\rangle, \tag{3}$$

in the Heisenberg picture of quantum mechanics, where the fermionic operators evolve accordingly to the Heisenberg equation. Therefore the rDM equation of motion reads

$$\frac{d}{dt}\rho^{\text{ee}}_{ab} = \left\langle \frac{i}{\hbar}[H, c^\dagger_b] c_a \right\rangle + \left\langle c^\dagger_b \frac{i}{\hbar}[H, c_a] \right\rangle. \tag{4}$$

where we have omitted the time dependency in the fermionic operators for compactness. This practice is done throughout the remaining work with no further mention. As shown in Appendix, given the equal-time fermionic anti-commutator properties $\left\{c_a, c_b^\dagger\right\} = \left\{c_b^\dagger, c_a\right\} = \delta_{ab}$ and $\{c_a, c_b\} = \left\{c_b^\dagger, c_a^\dagger\right\} = 0$, each of the commutators are evaluated to

$$[H, c_b^\dagger] = \bar{h}_{\alpha b} c_\alpha^\dagger + \frac{1}{4}\left(V_{\gamma b}^{\alpha\beta} - V_{b\gamma}^{\alpha\beta}\right)\left(c_\alpha^\dagger c_\beta^\dagger c_\gamma - c_\gamma c_\beta^\dagger c_\alpha^\dagger\right) \tag{5}$$

$$[H, c_a] = -\bar{h}_{a\alpha} c_\alpha + \frac{1}{4}\left(V_{\beta\gamma}^{\alpha a} - V_{\beta\gamma}^{a\alpha}\right)\left(c_\alpha^\dagger c_\beta c_\gamma - c_\gamma c_\beta c_\alpha^\dagger\right), \tag{6}$$

where we have hidden the summations on repeated indices using Einstein's notation.

Commutators calculations

First we calculate the commutator with $c_b^\dagger$ making use of the following commutator properties. We obtain

$$\left[c_\alpha^\dagger c_\beta, c_b^\dagger\right] = c_\alpha^\dagger \left\{c_\beta, c_b^\dagger\right\} - \cancel{\left\{c_b^\dagger, c_\alpha^\dagger\right\}} c_\beta = c_\alpha^\dagger \delta_{\beta b}$$
$$\left[c_\beta c_\alpha^\dagger, c_b^\dagger\right] = c_\beta \cancel{\left\{c_\alpha^\dagger, c_b^\dagger\right\}} - \left\{c_b^\dagger, c_\beta\right\} c_\alpha^\dagger = -c_\alpha^\dagger \delta_{\beta b}$$

and that

$$\left[c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta, c_b^\dagger\right] = c_\alpha^\dagger c_\beta^\dagger \left(c_\gamma \left\{c_\delta, c_b^\dagger\right\} - \left\{c_b^\dagger, c_\gamma\right\} c_\delta\right) + \left(c_\alpha^\dagger \cancel{\left\{c_\beta^\dagger, c_b^\dagger\right\}} - \cancel{\left\{c_b^\dagger, c_\alpha^\dagger\right\}} c_\beta^\dagger\right) c_\gamma c_\delta = c_\alpha^\dagger c_\beta^\dagger c_\gamma \delta_{\delta b} - c_\alpha^\dagger c_\beta^\dagger c_\delta \delta_{\gamma b}$$
$$\left[c_\delta c_\gamma c_\beta^\dagger c_\alpha^\dagger, c_b^\dagger\right] = c_\delta c_\gamma \left(c_\beta^\dagger \cancel{\left\{c_\alpha^\dagger, c_b^\dagger\right\}} - \cancel{\left\{c_b^\dagger, c_\beta^\dagger\right\}} c_\alpha^\dagger\right) + \left(c_\delta \left\{c_\gamma, c_b^\dagger\right\} - \left\{c_b^\dagger, c_\delta\right\} c_\gamma\right) c_\beta^\dagger c_\alpha^\dagger = c_\delta c_\beta^\dagger c_\alpha^\dagger \delta_{\gamma b} - c_\gamma c_\beta^\dagger c_\alpha^\dagger \delta_{\delta b}$$

The complete commutator then reads

$$\begin{aligned}
[H, c_b^\dagger] &= \left[\frac{1}{2}\bar{h}_{\alpha\beta}\left(c_\alpha^\dagger c_\beta - c_\beta c_\alpha^\dagger\right) + \frac{1}{4}V_{\gamma\delta}^{\alpha\beta}\left(c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta + c_\delta c_\gamma c_\beta^\dagger c_\alpha^\dagger\right) + \frac{1}{2}\cancel{\tilde{h}_0}, c_b^\dagger\right] \\
&= \frac{1}{2}\bar{h}_{\alpha\beta}\left(\left[c_\alpha^\dagger c_\beta, c_b^\dagger\right] - \left[c_\beta c_\alpha^\dagger, c_b^\dagger\right]\right) + \frac{1}{4}V_{\gamma\delta}^{\alpha\beta}\left(\left[c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta, c_b^\dagger\right] + \left[c_\delta c_\gamma c_\beta^\dagger c_\alpha^\dagger, c_b^\dagger\right]\right) \\
&= \bar{h}_{\alpha\beta} c_\alpha^\dagger \delta_{\beta b} + \frac{1}{4}\left(V_{\gamma\delta}^{\alpha\beta} c_\alpha^\dagger c_\beta^\dagger c_\gamma \delta_{\delta b} - V_{\gamma\delta}^{\alpha\beta} c_\alpha^\dagger c_\beta^\dagger c_\delta \delta_{\gamma b} + V_{\gamma\delta}^{\alpha\beta} c_\delta c_\beta^\dagger c_\alpha^\dagger \delta_{\gamma b} - V_{\gamma\delta}^{\alpha\beta} c_\gamma c_\beta^\dagger c_\alpha^\dagger \delta_{\delta b}\right) \\
&= \bar{h}_{\alpha b} c_\alpha^\dagger + \frac{1}{4}\left(V_{\gamma b}^{\alpha\beta} - V_{b\gamma}^{\alpha\beta}\right)\left(c_\alpha^\dagger c_\beta^\dagger c_\gamma - c_\gamma c_\beta^\dagger c_\alpha^\dagger\right)
\end{aligned}$$

Analogous for the $[H, c_a]$ commutator, we have

$$\left[c_\alpha^\dagger c_\beta, c_a\right] = c_\alpha^\dagger \cancel{\{c_\beta, c_a\}} - \left\{c_a, c_\alpha^\dagger\right\} c_\beta = -c_\beta \delta_{a\alpha}$$
$$\left[c_\beta c_\alpha^\dagger, c_a\right] = c_\beta \left\{c_\alpha^\dagger, c_a\right\} - \cancel{\{c_a, c_\beta\}} c_\alpha^\dagger = c_\beta \delta_{a\alpha}$$

and that

$$\left[c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta, c_a\right] = c_\alpha^\dagger c_\beta^\dagger \left(c_\gamma \cancel{\{c_\delta, c_a\}} - \cancel{\{c_a, c_\gamma\}} c_\delta\right) + \left(c_\alpha^\dagger \left\{c_\beta^\dagger, c_a\right\} - \left\{c_a, c_\alpha^\dagger\right\} c_\beta^\dagger\right) c_\gamma c_\delta = c_\alpha^\dagger c_\gamma c_\delta \delta_{\beta a} - c_\beta^\dagger c_\gamma c_\delta \delta_{a\alpha}$$
$$\left[c_\delta c_\gamma c_\beta^\dagger c_\alpha^\dagger, c_a\right] = c_\delta c_\gamma \left(c_\beta^\dagger \left\{c_\alpha^\dagger, c_a\right\} - \left\{c_a, c_\beta^\dagger\right\} c_\alpha^\dagger\right) + \left(c_\delta \cancel{\{c_\gamma, c_a\}} - \cancel{\{c_a, c_\delta\}} c_\gamma\right) c_\beta^\dagger c_\alpha^\dagger = c_\delta c_\gamma c_\beta^\dagger \delta_{\alpha a} - c_\delta c_\gamma c_\alpha^\dagger \delta_{a\beta}$$

The complete commutator then reads

$$\begin{aligned}
[H, c_a] &= \left[\frac{1}{2}\bar{h}_{\alpha\beta}\left(c_\alpha^\dagger c_\beta - c_\beta c_\alpha^\dagger\right) + \frac{1}{4}V_{\gamma\delta}^{\alpha\beta}\left(c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta + c_\delta c_\gamma c_\beta^\dagger c_\alpha^\dagger\right) + \frac{1}{2}\cancel{\tilde{h}_0}, c_a\right] \\
&= \frac{1}{2}\bar{h}_{\alpha\beta}\left(\left[c_\alpha^\dagger c_\beta, c_a\right] - \left[c_\beta c_\alpha^\dagger, c_a\right]\right) + \frac{1}{4}V_{\gamma\delta}^{\alpha\beta}\left(\left[c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta, c_a\right] + \left[c_\delta c_\gamma c_\beta^\dagger c_\alpha^\dagger, c_a\right]\right) \\
&= -\bar{h}_{\alpha\beta} c_\beta \delta_{a\alpha} + \frac{1}{4}\left(V_{\gamma\delta}^{\alpha\beta} c_\alpha^\dagger c_\gamma c_\delta \delta_{\beta a} - V_{\gamma\delta}^{\alpha\beta} c_\beta^\dagger c_\gamma c_\delta \delta_{a\alpha} + V_{\gamma\delta}^{\alpha\beta} c_\delta c_\gamma c_\beta^\dagger \delta_{\alpha a} - V_{\gamma\delta}^{\alpha\beta} c_\delta c_\gamma c_\alpha^\dagger \delta_{a\beta}\right) \\
&= -\bar{h}_{a\beta} c_\beta + \frac{1}{4}\left(V_{\gamma\delta}^{\alpha a} c_\alpha^\dagger c_\gamma c_\delta - V_{\gamma\delta}^{a\beta} c_\beta^\dagger c_\gamma c_\delta + V_{\gamma\delta}^{a\beta} c_\delta c_\gamma c_\beta^\dagger - V_{\gamma\delta}^{\alpha a} c_\delta c_\gamma c_\alpha^\dagger\right) \\
&= -\bar{h}_{a\alpha} c_\alpha + \frac{1}{4}\left(V_{\beta\gamma}^{\alpha a} - V_{\beta\gamma}^{a\alpha}\right)\left(c_\alpha^\dagger c_\beta c_\gamma - c_\gamma c_\beta c_\alpha^\dagger\right)
\end{aligned}$$

$$\{c_a, c_b^\dagger\} = \{c_b^\dagger, c_a\} = \delta_{ab}$$
$$\{c_a, c_b\} = \{c_b^\dagger, c_a^\dagger\} = 0$$

$$[AB, Z] = A\{B, Z\} - \{Z, A\}B$$
$$[AB, Z] = A[B, Z] - [Z, A]B$$
$$[AB, CD] = -A(C\{D, B\} - \{B, C\}D) - (C\{D, A\} - \{A, C\}D)B$$
$$[ABCD, Z] = AB(C\{D, Z\} - \{Z, C\}D) + (A\{B, Z\} - \{Z, A\}B)CD$$

Substituting Eqs.(5) and (6) back into Eq.(4) yields

$$\begin{aligned} -i\hbar\frac{d}{dt}\rho_{ab}^{\mathrm{ee}} &= \rho_{a\alpha}^{\mathrm{ee}}\bar{h}_{\alpha b} - \bar{h}_{a\alpha}\rho_{\alpha b}^{\mathrm{ee}} \\ &+ \frac{1}{4}\left(V_{\gamma b}^{\alpha\beta} - V_{b\gamma}^{\alpha\beta}\right)\left(\left\langle c_\alpha^\dagger c_\beta^\dagger c_\gamma c_a\right\rangle - \left\langle c_\gamma c_\beta^\dagger c_\alpha^\dagger c_a\right\rangle\right) \\ &+ \frac{1}{4}\left(V_{\beta\gamma}^{\alpha a} - V_{\beta\gamma}^{a\alpha}\right)\left(\left\langle c_b^\dagger c_\alpha^\dagger c_\beta c_\gamma\right\rangle - \left\langle c_b^\dagger c_\gamma c_\beta c_\alpha^\dagger\right\rangle\right) \end{aligned} \tag{7}$$

where we came back to the definition of the normal rDM in Eq.(**??**) for the first two terms. Notice, however, that the remaining interaction terms correspond to the expectation values of four-operators. For this, we introduce a mean-field approximation where we assume the two-particle expectation value to simply behave as a product of two one-particle expectation values. From this mean-field decoupling we can then make use of *Wick's theorem* as done in the Appendix. We obtain

$$\left\langle c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta\right\rangle \approx \rho_{\beta\alpha}^{\mathrm{he}}\rho_{\delta\gamma}^{\mathrm{eh}} - \rho_{\gamma\alpha}^{\mathrm{ee}}\rho_{\delta\beta}^{\mathrm{hh}} + \rho_{\delta\alpha}^{\mathrm{ee}}\rho_{\gamma\beta}^{\mathrm{ee}} \tag{8}$$

$$\left\langle c_\alpha c_\beta^\dagger c_\gamma^\dagger c_\delta\right\rangle \approx \rho_{\beta\alpha}^{\mathrm{hh}}\rho_{\delta\gamma}^{\mathrm{ee}} - \rho_{\gamma\alpha}^{\mathrm{hh}}\rho_{\delta\beta}^{\mathrm{ee}} + \rho_{\delta\alpha}^{\mathrm{eh}}\rho_{\gamma\beta}^{\mathrm{he}} \tag{9}$$

$$\left\langle c_\alpha^\dagger c_\beta c_\gamma c_\delta^\dagger\right\rangle \approx \rho_{\beta\alpha}^{\mathrm{ee}}\rho_{\delta\gamma}^{\mathrm{ee}} - \rho_{\gamma\alpha}^{\mathrm{ee}}\rho_{\delta\beta}^{\mathrm{hh}} + \rho_{\delta\alpha}^{\mathrm{he}}\rho_{\gamma\beta}^{\mathrm{eh}} \tag{10}$$

As derived below in the gray appendices, the purely electronic rDM equation of motion can then be written in a time-dependent Hartree-Fock in a more compact and elegant manner as

$$i\hbar\frac{d}{dt}\rho_{ab}^{\mathrm{ee}} = \left[h_{ab}^{\mathrm{HF}}, \rho_{ab}^{\mathrm{ee}}\right] - \rho_{ab}^{\mathrm{ee}}\Delta_{ba}^* + \Delta_{ab}\rho_{ab}^{\mathrm{he}} \tag{11}$$

with the Hartree-Fock Hamiltonian being

$$h_{ab}^{HF}(\rho) = h_{ab} + \Sigma_{ab}^H(\rho) + \Sigma_{ab}^F(\rho), \tag{12}$$

identifying the Hartree, Fock and pairing self-energies (themselves dependent on the rDM) respectively as

$$\begin{aligned} \Sigma_{ab}^H &= \textstyle\sum_{\gamma\delta} V_{b\gamma}^{\delta a}\rho_{\gamma\delta}^{\mathrm{ee}} \\ \Sigma_{ab}^F &= -\textstyle\sum_{\gamma\delta} V_{b\gamma}^{a\delta}\rho_{\gamma\delta}^{\mathrm{ee}} \end{aligned} \quad \text{and} \quad \Delta_{ab} = \sum_{\gamma\delta} V_{\delta\gamma}^{ab}\rho_{\gamma\delta}^{\mathrm{eh}} \tag{13}$$

As a side note concerning excitons, notice that the Fock self-energy term in is an attractive interaction (due the negative sign) such that it lower the total energy of the system enabling the formation of the electron-hole bound states. Conversely, the Hartree self-energy term will control details of the excitation spectrum, such as spin-splitting. Also, note the symmetry $\Delta_{ab} = -\Delta_{ba}$.

Purely electronic rDM EoM calculations

$$[\mathcal{H}, c_b^\dagger] = h_{\alpha b}c_\alpha^\dagger + \frac{1}{2}(V_{\gamma b}^{\alpha\beta} - V_{b\gamma}^{\alpha\beta})c_\alpha^\dagger c_\beta^\dagger c_\gamma$$
$$[\mathcal{H}, c_a] = -h_{a\alpha}c_\alpha + \frac{1}{2}(V_{\beta\gamma}^{\alpha a} - V_{\beta\gamma}^{a\alpha})c_\alpha^\dagger c_\beta c_\gamma$$

Using the mean-field decoupling in Eq.(**??**), the *normal* EoM interaction terms becomes

$$
\begin{aligned}
&-\frac{1}{2}(V^{\alpha\beta}_{\gamma b}-V^{\alpha\beta}_{b\gamma})(\rho^{\mathrm{he}}_{\beta\alpha}\rho^{\mathrm{eh}}_{a\gamma}-\rho^{\mathrm{ee}}_{\gamma\alpha}\rho^{\mathrm{ee}}_{a\beta}+\rho^{\mathrm{ee}}_{a\alpha}\rho^{\mathrm{ee}}_{\gamma\beta})\\
&-\frac{1}{2}(V^{\alpha a}_{\beta\gamma}-V^{a\alpha}_{\beta\gamma})(\rho^{\mathrm{he}}_{\alpha b}\rho^{\mathrm{eh}}_{\gamma\beta}-\rho^{\mathrm{ee}}_{\beta b}\rho^{\mathrm{ee}}_{\gamma\alpha}+\rho^{\mathrm{ee}}_{\gamma b}\rho^{\mathrm{ee}}_{\beta\alpha})
\end{aligned}
$$

with the purely electronic reading

$$
\begin{aligned}
&-\rho^{\mathrm{ee}}_{a\beta}(V^{\alpha\beta}_{b\gamma}-V^{\beta\alpha}_{b\gamma})\rho^{\mathrm{ee}}_{\gamma\alpha}+(V^{\alpha a}_{\beta\gamma}-V^{a\alpha}_{\beta\gamma})\rho^{\mathrm{ee}}_{\gamma\alpha}\rho^{\mathrm{ee}}_{\beta b}\\
&=-\rho^{\mathrm{ee}}_{a\beta}\Sigma^{\mathrm{HF}}_{\beta b}+\Sigma^{\mathrm{HF}}_{a\beta}\rho^{\mathrm{ee}}_{\beta b}=[\Sigma_{\mathrm{HF}},\rho_{\mathrm{ee}}]|_{ab}
\end{aligned}
$$

(no index is a matrix since I reserve the bold specifically to denote the spin structure) and the anomalous terms reads

$$
\begin{aligned}
&-\frac{1}{2}\rho^{\mathrm{eh}}_{a\gamma}(V^{\beta\alpha}_{b\gamma}-V^{\beta\alpha}_{\gamma b})(\rho^{\mathrm{eh}}_{\alpha\beta})^{*}-\frac{1}{2}(V^{\alpha a}_{\beta\gamma}-V^{a\alpha}_{\beta\gamma})\rho^{\mathrm{eh}}_{\gamma\beta}\rho^{\mathrm{he}}_{\alpha b}\\
=&-\frac{1}{2}\rho^{\mathrm{eh}}_{a\gamma}(\Delta_{b\gamma}-\Delta_{\gamma b})^{*}-\frac{1}{2}(\Delta_{\alpha a}-\Delta_{a\alpha})\rho^{\mathrm{he}}_{\alpha b}\\
=&\,\rho_{\mathrm{eh}}\cdot\Delta^{*}|_{ab}+\Delta\cdot\rho_{\mathrm{he}}|_{ab}
\end{aligned}
$$

The non-interacting terms is trivially $-[h,\rho_{\mathrm{ee}}]_{ab}$. Analogously for the *anomalous* EoM, we have trivially non-interacting term $-[\rho_{\mathrm{eh}}\cdot h^{T}]_{ab}-[h\cdot\rho_{\mathrm{eh}}]_{ab}$. For the interaction terms we perform instead the mean-field decoupling of $\langle c^{\dagger}_{\alpha}c_{\beta}c_{\gamma}c_{a}\rangle$ and $\langle c_{b}c^{\dagger}_{\alpha}c_{\beta}c_{\gamma}\rangle$, we obtain

$$
\begin{aligned}
&-\frac{1}{2}(V^{\alpha b}_{\beta\gamma}-V^{b\alpha}_{\beta\gamma})(\rho^{\mathrm{ee}}_{\beta\alpha}\rho^{\mathrm{eh}}_{a\gamma}-\rho^{\mathrm{ee}}_{\gamma\alpha}\rho^{\mathrm{eh}}_{a\beta}+\rho^{\mathrm{ee}}_{a\alpha}\rho^{\mathrm{eh}}_{\gamma\beta})\\
&-\frac{1}{2}(V^{\alpha a}_{\beta\gamma}-V^{a\alpha}_{\beta\gamma})(\rho^{\mathrm{hh}}_{\alpha b}\rho^{\mathrm{eh}}_{\gamma\beta}-\rho^{\mathrm{eh}}_{\beta b}\rho^{\mathrm{ee}}_{\gamma\alpha}+\rho^{\mathrm{eh}}_{\gamma b}\rho^{\mathrm{ee}}_{\beta\alpha})
\end{aligned}
$$

The first term reading

$$
\begin{aligned}
&\rho^{\mathrm{eh}}_{a\gamma}(V^{\alpha b}_{\gamma\beta}-V^{b\alpha}_{\gamma\beta})\rho^{\mathrm{ee}}_{\beta\alpha}-\frac{1}{2}\rho^{\mathrm{ee}}_{a\alpha}(V^{\alpha b}_{\beta\gamma}-V^{b\alpha}_{\beta\gamma})\rho^{\mathrm{eh}}_{\gamma\beta}\\
=&\rho^{\mathrm{eh}}_{a\gamma}\Sigma^{\mathrm{HF}}_{b\gamma}-\frac{1}{2}\rho^{\mathrm{ee}}_{a\alpha}(\Delta_{\alpha b}-\Delta_{b\alpha})\\
=&\,\rho_{\mathrm{eh}}\cdot\Sigma^{*}_{\mathrm{HF}}|_{ab}-\rho_{\mathrm{ee}}\cdot\Delta|_{ab}
\end{aligned}
$$

and the second one reading

$$
\begin{aligned}
&-\frac{1}{2}\left(V^{\alpha a}_{\beta\gamma}-V^{a\alpha}_{\beta\gamma}\right)\rho^{\mathrm{eh}}_{\gamma\beta}\rho^{\mathrm{hh}}_{\alpha b}+\left(V^{\alpha a}_{\beta\gamma}-V^{a\alpha}_{\beta\gamma}\right)\rho^{\mathrm{ee}}_{\gamma\alpha}\rho^{\mathrm{eh}}_{\beta b}\\
=&-\frac{1}{2}\left(\Delta_{\alpha a}-\Delta_{a\alpha}\right)\rho^{\mathrm{hh}}_{\alpha b}+\Sigma^{\mathrm{HF}}_{a\beta}\rho^{\mathrm{eh}}_{\beta b}\\
=&\,\Delta\cdot\rho_{\mathrm{hh}}|_{ab}+\Sigma_{\mathrm{HF}}\cdot\rho_{\mathrm{eh}}|_{ab}
\end{aligned}
$$

Confirming that the result actually checks out

$$
\begin{aligned}
&\begin{pmatrix} h^{\mathrm{HF}}_{ab} & \Delta_{ab}\\ \Delta^{*}_{ba} & -h^{\mathrm{HF}}_{ba}\end{pmatrix}\begin{pmatrix}\rho^{\mathrm{ee}}_{ab} & \rho^{\mathrm{eh}}_{ab}\\ \rho^{\mathrm{he}}_{ab} & \rho^{\mathrm{hh}}_{ab}\end{pmatrix}-\begin{pmatrix}\rho^{\mathrm{ee}}_{ab} & \rho^{\mathrm{eh}}_{ab}\\ \rho^{\mathrm{he}}_{ab} & \rho^{\mathrm{hh}}_{ab}\end{pmatrix}\begin{pmatrix} h^{\mathrm{HF}}_{ab} & \Delta_{ab}\\ \Delta^{*}_{ba} & -h^{\mathrm{HF}}_{ba}\end{pmatrix}\\
=&\begin{bmatrix}\left(h^{\mathrm{HF}}_{ab}\rho^{\mathrm{ee}}_{ab}+\Delta_{ab}\rho^{\mathrm{he}}_{ab}\right) & \left(h^{\mathrm{HF}}_{ab}\rho^{\mathrm{eh}}_{ab}+\Delta_{ab}\rho^{\mathrm{hh}}_{ab}\right)\\ \circ & \circ\end{bmatrix}-\begin{bmatrix}\left(\rho^{\mathrm{ee}}_{ab}h^{\mathrm{HF}}_{ab}-\rho^{\mathrm{eh}}_{ab}\Delta^{*}_{ba}\right) & \left(\rho^{\mathrm{ee}}_{ab}\Delta_{ab}-\rho^{\mathrm{eh}}_{ab}h^{\mathrm{HF}}_{ba}\right)\\ \circ & \circ\end{bmatrix}\\
=&\begin{bmatrix}\left([h^{\mathrm{HF}}_{ab},\rho^{\mathrm{ee}}_{ab}]+\Delta_{ab}\rho_{\mathrm{he}}+\rho^{\mathrm{eh}}_{ab}\Delta^{*}_{ba}\right) & \left(h^{\mathrm{HF}}_{ab}\rho^{\mathrm{eh}}_{ab}-\rho^{\mathrm{eh}}_{ab}h^{\mathrm{HF}}_{ba}-\rho^{\mathrm{ee}}_{ab}\Delta_{ab}+\Delta_{ab}\rho^{\mathrm{hh}}_{ab}\right)\\ \circ & \circ\end{bmatrix}
\end{aligned}
$$

$$\rho^{\mathrm{hh}}_{ab} = \left\langle c_b c_a^\dagger \right\rangle = -\left\langle c_a^\dagger c_b \right\rangle + \delta_{ab} = -\rho^{\mathrm{ee}}_{ba} + \delta_{ab}$$
$$\rho^{\mathrm{he}}_{ab} = \left\langle c_b^\dagger c_a^\dagger \right\rangle = \left\langle c_a c_b \right\rangle^\dagger = \left(\rho^{\mathrm{eh}}_{ba}\right)^\dagger$$
$$\{c_a, c_b\} = 0 \Leftrightarrow c_a c_b = -c_b c_a \Rightarrow \rho^{\mathrm{eh}}_{ba} = -\rho^{\mathrm{eh}}_{ab}$$
$$\left\{c_a^\dagger, c_b^\dagger\right\} = 0 \Leftrightarrow c_a^\dagger c_b^\dagger = -c_b^\dagger c_a^\dagger \Rightarrow \rho^{\mathrm{eh}}_{ba} = -\rho^{\mathrm{eh}}_{ab}$$
$$\Delta_{ab} = V^{ab}_{\delta\gamma}\rho^{\mathrm{eh}}_{\gamma\delta} = V^{ba}_{\gamma\delta}\rho^{\mathrm{eh}}_{\gamma\delta} \overset{\delta \rightleftarrows \gamma}{=} V^{ba}_{\delta\gamma}\rho^{\mathrm{eh}}_{\delta\gamma} = V^{ba}_{\delta\gamma}\left(-\rho^{\mathrm{eh}}_{\gamma\delta}\right) = -\Delta_{ba}$$

Analogously for the anomalous EoM we have that

$$\frac{d}{dt}\rho^{\mathrm{eh}}_{ab} = \left\langle \frac{i}{\hbar}[H, c_b]c_a \right\rangle - \left\langle c_b \frac{i}{\hbar}[H, c_a] \right\rangle. \tag{14}$$

but instead perform the mean-field decoupling

$$\left\langle c_\alpha^\dagger c_\beta c_\gamma c_\delta \right\rangle \approx \rho^{\mathrm{ee}}_{\beta\alpha}\rho^{\mathrm{eh}}_{\delta\gamma} - \rho^{\mathrm{eh}}_{\gamma\alpha}\rho^{\mathrm{eh}}_{\delta\beta} + \rho^{\mathrm{ee}}_{\delta\alpha}\rho^{\mathrm{eh}}_{\gamma\beta}, \tag{15}$$
$$\left\langle c_\alpha c_\beta^\dagger c_\gamma c_\delta \right\rangle \approx \rho^{\mathrm{he}}_{\beta\alpha}\rho^{\mathrm{eh}}_{\delta\gamma} - \rho^{\mathrm{eh}}_{\gamma\alpha}\rho^{\mathrm{ee}}_{\delta\beta} + \rho^{\mathrm{eh}}_{\delta\alpha}\rho^{\mathrm{ee}}_{\gamma\beta}. \tag{16}$$
$$\left\langle c_\alpha c_\beta c_\gamma^\dagger c_\delta \right\rangle \approx \rho^{\mathrm{eh}}_{\beta\alpha}\rho^{\mathrm{ee}}_{\delta\gamma} - \rho^{\mathrm{hh}}_{\gamma\alpha}\rho^{\mathrm{eh}}_{\delta\beta} + \rho^{\mathrm{eh}}_{\delta\alpha}\rho^{\mathrm{hh}}_{\gamma\beta} \tag{17}$$
$$\left\langle c_\alpha c_\beta c_\gamma c_\delta^\dagger \right\rangle \approx \rho^{\mathrm{eh}}_{\beta\alpha}\rho^{\mathrm{hh}}_{\delta\gamma} - \rho^{\mathrm{eh}}_{\gamma\alpha}\rho^{\mathrm{hh}}_{\delta\beta} + \rho^{\mathrm{hh}}_{\delta\alpha}\rho^{\mathrm{eh}}_{\gamma\beta} \tag{18}$$

which yields the anomalous equation of motion

$$i\hbar\frac{d}{dt}\boldsymbol{\rho}_{\mathrm{eh}} = \mathbf{H}_{\mathrm{HF}}\boldsymbol{\rho}_{\mathrm{eh}} + \boldsymbol{\rho}_{\mathrm{eh}}\mathbf{H}_{\mathrm{HF}}^T + \boldsymbol{\Delta}\boldsymbol{\rho}_{\mathrm{hh}} - \boldsymbol{\rho}_{\mathrm{ee}}\boldsymbol{\Delta} \tag{19}$$

Anomalous rDM EoM calculations
Using the result shown in Eq.(6), the EoM in Eq.(14) reads

$$\begin{aligned} i\hbar\frac{d}{dt}\rho^{\mathrm{eh}}_{ab} =& -\bar{h}_{b\alpha}\rho^{\mathrm{eh}}_{a\alpha} + \frac{1}{4}\left(V^{\alpha b}_{\beta\gamma} - V^{b\alpha}_{\beta\gamma}\right)\left(\left\langle c_\alpha^\dagger c_\beta c_\gamma c_a \right\rangle - \left\langle c_\gamma c_\beta c_\alpha^\dagger c_a \right\rangle\right) \\ & - \bar{h}_{a\alpha}\rho^{\mathrm{eh}}_{\alpha b} + \frac{1}{4}\left(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma}\right)\left(\left\langle c_b c_\alpha^\dagger c_\beta c_\gamma \right\rangle - \left\langle c_b c_\gamma c_\beta c_\alpha^\dagger \right\rangle\right) \end{aligned}$$

Immediately, the single particle term reads

$$-\bar{h}_{b\alpha}\rho^{\mathrm{eh}}_{a\alpha} - \bar{h}_{a\alpha}\rho^{\mathrm{eh}}_{\alpha b} = -\boldsymbol{\rho}_{\mathrm{eh}}\bar{\boldsymbol{h}}^T - \bar{\boldsymbol{h}}\boldsymbol{\rho}_{\mathrm{eh}}$$

From Wick's theorem we have that

$$\begin{aligned} & \frac{1}{4}\left(V^{\alpha b}_{\beta\gamma} - V^{b\alpha}_{\beta\gamma}\right)\left(\left[\rho^{\mathrm{ee}}_{\beta\alpha}\rho^{\mathrm{eh}}_{a\gamma} - \rho^{\mathrm{ee}}_{\gamma\alpha}\rho^{\mathrm{eh}}_{a\beta} + \rho^{\mathrm{ee}}_{a\alpha}\rho^{\mathrm{eh}}_{\gamma\beta}\right] - \left[\rho^{\mathrm{eh}}_{\beta\gamma}\rho^{\mathrm{ee}}_{a\alpha} - \rho^{\mathrm{hh}}_{\alpha\gamma}\rho^{\mathrm{eh}}_{a\beta} + \rho^{\mathrm{eh}}_{a\gamma}\rho^{\mathrm{hh}}_{\alpha\beta}\right]\right) \\ +& \frac{1}{4}\left(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma}\right)\left(\left[\rho^{\mathrm{hh}}_{\alpha b}\rho^{\mathrm{eh}}_{\gamma\beta} - \rho^{\mathrm{eh}}_{\beta b}\rho^{\mathrm{ee}}_{\gamma\alpha} + \rho^{\mathrm{eh}}_{\gamma b}\rho^{\mathrm{ee}}_{\beta\alpha}\right] - \left[\rho^{\mathrm{eh}}_{\gamma b}\rho^{\mathrm{hh}}_{\alpha\beta} - \rho^{\mathrm{eh}}_{\beta b}\rho^{\mathrm{hh}}_{\alpha\gamma} + \rho^{\mathrm{hh}}_{\alpha b}\rho^{\mathrm{eh}}_{\beta\gamma}\right]\right) \end{aligned}$$

Let us start by focusing only on the first two terms,

$$\begin{aligned} =& \frac{1}{4}\left(V^{\alpha b}_{\beta\gamma} - V^{b\alpha}_{\beta\gamma}\right)\rho^{\mathrm{ee}}_{\beta\alpha}\rho^{\mathrm{eh}}_{a\gamma} - \frac{1}{4}\left(V^{\alpha b}_{\beta\gamma} - V^{b\alpha}_{\beta\gamma}\right)\rho^{\mathrm{ee}}_{\gamma\alpha}\rho^{\mathrm{eh}}_{a\beta} + \frac{1}{4}\left(V^{\alpha b}_{\beta\gamma} - V^{b\alpha}_{\beta\gamma}\right)\rho^{\mathrm{ee}}_{a\alpha}\rho^{\mathrm{eh}}_{\gamma\beta} \\ & - \frac{1}{4}\left(V^{\alpha b}_{\beta\gamma} - V^{b\alpha}_{\beta\gamma}\right)\rho^{\mathrm{eh}}_{\beta\gamma}\rho^{\mathrm{ee}}_{a\alpha} + \frac{1}{4}\left(V^{\alpha b}_{\beta\gamma} - V^{b\alpha}_{\beta\gamma}\right)\rho^{\mathrm{hh}}_{\alpha\gamma}\rho^{\mathrm{eh}}_{a\beta} - \frac{1}{4}\left(V^{\alpha b}_{\beta\gamma} - V^{b\alpha}_{\beta\gamma}\right)\rho^{\mathrm{eh}}_{a\gamma}\rho^{\mathrm{hh}}_{\alpha\beta} \\ =& \frac{1}{2}\rho^{\mathrm{eh}}_{a\gamma}\left[\left(-V^{\alpha b}_{\gamma\beta} + V^{b\alpha}_{\gamma\beta}\right)\rho^{\mathrm{ee}}_{\beta\alpha} + \left(V^{\alpha b}_{\gamma\beta} - V^{b\alpha}_{\gamma\beta}\right)\left(-\rho^{\mathrm{ee}}_{\beta\alpha} + \delta_{\alpha\beta}\right)\right] - \frac{1}{2}\rho^{\mathrm{ee}}_{a\alpha}\left(V^{\alpha b}_{\beta\gamma} - V^{b\alpha}_{\beta\gamma}\right)\rho^{\mathrm{eh}}_{\beta\gamma} \\ =& \rho^{\mathrm{eh}}_{a\gamma}\left(-\Sigma^H_{b\gamma} - \Sigma^F_{b\gamma}\right) + \rho^{\mathrm{eh}}_{a\gamma}\mu_{b\gamma} - \frac{1}{2}\rho^{\mathrm{ee}}_{a\alpha}\left(\Delta_{b\alpha} - \Delta_{\alpha b}\right) \\ =& \boldsymbol{\rho}_{\mathrm{eh}}\left(-\boldsymbol{\Sigma}^T + \boldsymbol{\mu}^T\right) + \boldsymbol{\rho}_{\mathrm{ee}}\boldsymbol{\Delta} \end{aligned}$$

and next on the terms two terms

$$
\begin{aligned}
&\frac{1}{4}\left(V^{\alpha a}_{\beta\gamma}-V^{a\alpha}_{\beta\gamma}\right)\rho^{\mathrm{hh}}_{\alpha b}\rho^{\mathrm{eh}}_{\gamma\beta}-\frac{1}{4}\left(V^{\alpha a}_{\beta\gamma}-V^{a\alpha}_{\beta\gamma}\right)\rho^{\mathrm{eh}}_{\beta b}\rho^{\mathrm{ee}}_{\gamma\alpha}+\frac{1}{4}\left(V^{\alpha a}_{\beta\gamma}-V^{a\alpha}_{\beta\gamma}\right)\rho^{\mathrm{eh}}_{\gamma b}\rho^{\mathrm{ee}}_{\beta\alpha}\\
&-\frac{1}{4}\left(V^{\alpha a}_{\beta\gamma}-V^{a\alpha}_{\beta\gamma}\right)\rho^{\mathrm{eh}}_{\gamma b}\rho^{\mathrm{hh}}_{\alpha\beta}+\frac{1}{4}\left(V^{\alpha a}_{\beta\gamma}-V^{a\alpha}_{\beta\gamma}\right)\rho^{\mathrm{eh}}_{\beta b}\rho^{\mathrm{hh}}_{\alpha\gamma}-\frac{1}{4}\left(V^{\alpha a}_{\beta\gamma}-V^{a\alpha}_{\beta\gamma}\right)\rho^{\mathrm{hh}}_{\alpha b}\rho^{\mathrm{eh}}_{\beta\gamma}\\
=&\frac{1}{2}\left(V^{\alpha a}_{\beta\gamma}-V^{a\alpha}_{\beta\gamma}\right)\rho^{\mathrm{eh}}_{\gamma\beta}\rho^{\mathrm{hh}}_{\alpha b}+\frac{1}{2}\left(-V^{\alpha a}_{\beta\gamma}+V^{a\alpha}_{\beta\gamma}\right)\left(-\rho^{\mathrm{ee}}_{\beta\alpha}+\delta_{\alpha\beta}\right)\rho^{\mathrm{eh}}_{\gamma b}+\frac{1}{2}\left(-V^{\alpha a}_{\gamma\beta}+V^{a\alpha}_{\gamma\beta}\right)\rho^{\mathrm{ee}}_{\beta\alpha}\rho^{\mathrm{eh}}_{\gamma b}\\
=&\frac{1}{2}\left(\Delta_{\alpha a}-\Delta_{a\alpha}\right)\rho^{\mathrm{hh}}_{\alpha b}+\frac{1}{2}\left(-\Sigma^{F}_{a\gamma}-\Sigma^{H}_{a\gamma}\right)\rho^{\mathrm{eh}}_{\gamma b}+\mu_{a\gamma}\rho^{\mathrm{eh}}_{\gamma b}+\frac{1}{2}\left(-\Sigma^{H}_{a\gamma}-\Sigma^{F}_{a\gamma}\right)\rho^{\mathrm{eh}}_{\gamma b}\\
=&-\Delta_{a\alpha}\rho^{\mathrm{hh}}_{\alpha b}-\Sigma_{a\gamma}\rho^{\mathrm{eh}}_{\gamma b}+\mu_{a\gamma}\rho^{\mathrm{eh}}_{\gamma b}\\
=&\left(-\boldsymbol{\Sigma}+\boldsymbol{\mu}\right)\boldsymbol{\rho}_{\mathrm{eh}}-\boldsymbol{\Delta}\boldsymbol{\rho}_{\mathrm{hh}}
\end{aligned}
$$

Putting these three terms together we obtain

$$
\begin{aligned}
&\left[\boldsymbol{\rho}_{\mathrm{eh}}\left(-\boldsymbol{\Sigma}^{T}+\boldsymbol{\mu}^{T}\right)+\boldsymbol{\rho}_{\mathrm{ee}}\boldsymbol{\Delta}\right]+\left[\left(-\boldsymbol{\Sigma}+\boldsymbol{\mu}\right)\boldsymbol{\rho}_{\mathrm{eh}}-\boldsymbol{\Delta}\boldsymbol{\rho}_{\mathrm{hh}}\right]+\left[-\boldsymbol{\rho}_{\mathrm{eh}}\bar{\boldsymbol{h}}^{T}-\bar{\boldsymbol{h}}\boldsymbol{\rho}_{\mathrm{eh}}\right]\\
=&-\left(\left(\bar{\boldsymbol{h}}-\boldsymbol{\mu}\right)+\boldsymbol{\Sigma}\right)\boldsymbol{\rho}_{\mathrm{eh}}-\boldsymbol{\rho}_{\mathrm{eh}}\left(\left(\bar{\boldsymbol{h}}-\boldsymbol{\mu}\right)+\boldsymbol{\Sigma}\right)^{T}-\boldsymbol{\Delta}\boldsymbol{\rho}_{\mathrm{hh}}+\boldsymbol{\rho}_{\mathrm{ee}}\boldsymbol{\Delta}\\
=&-\mathbf{H}_{\mathrm{HF}}\boldsymbol{\rho}_{\mathrm{eh}}-\boldsymbol{\rho}_{\mathrm{eh}}\mathbf{H}^{T}_{\mathrm{HF}}-\boldsymbol{\Delta}\boldsymbol{\rho}_{\mathrm{hh}}+\boldsymbol{\rho}_{\mathrm{ee}}\boldsymbol{\Delta}
\end{aligned}
$$

Finally, the complete reduced density matrix equation of motions yields

$$
i\hbar\frac{d}{dt}\boldsymbol{\rho}=\left[\mathbf{H}_{\mathrm{BdG}},\boldsymbol{\rho}\right] \tag{20}
$$

where we rewrote this expression in terms of an effective Nambu Hamiltonian $\mathbf{H}_{\mathrm{BdG}}$ defined as

$$
\mathbf{H}_{\mathrm{BdG}}=\begin{bmatrix}\mathbf{H}_{\mathrm{HF}} & \boldsymbol{\Delta}\\ \boldsymbol{\Delta}^{\dagger} & -\mathbf{H}^{T}_{\mathrm{HF}}\end{bmatrix} \tag{21}
$$

Furthermore, using the relations initially shown in Eqs.(**??**), one can express both the EoMs in Eq.(11) and Eq.(19) only in terms of the electronic normal $\boldsymbol{\rho}_{\mathrm{n}}\equiv\boldsymbol{\rho}_{\mathrm{ee}}$ and anomalous $\boldsymbol{\rho}_{\mathrm{a}}\equiv\boldsymbol{\rho}_{\mathrm{eh}}$ rDMs (instead of the four distinct terms $\boldsymbol{\rho}_{\mathrm{ee}}$, $\boldsymbol{\rho}_{\mathrm{eh}}$, $\boldsymbol{\rho}_{\mathrm{he}}$ and $\boldsymbol{\rho}_{\mathrm{hh}}$). We write

$$
\begin{aligned}
i\hbar\frac{d}{dt}\boldsymbol{\rho}_{\mathrm{n}}=&\left[\mathbf{H}_{\mathrm{HF}},\boldsymbol{\rho}_{\mathrm{n}}\right]-\left(\boldsymbol{\rho}_{\mathrm{a}}\boldsymbol{\Delta}^{\dagger}+\left(\boldsymbol{\rho}_{\mathrm{a}}\boldsymbol{\Delta}^{\dagger}\right)^{\dagger}\right)\\
i\hbar\frac{d}{dt}\boldsymbol{\rho}_{\mathrm{a}}=&\mathbf{H}_{\mathrm{HF}}\boldsymbol{\rho}_{\mathrm{a}}+\boldsymbol{\rho}_{\mathrm{a}}\mathbf{H}^{T}_{\mathrm{HF}}-\left(\boldsymbol{\Delta}\boldsymbol{\rho}^{\dagger}_{\mathrm{n}}+\boldsymbol{\rho}_{\mathrm{n}}\boldsymbol{\Delta}\right)+\boldsymbol{\Delta}
\end{aligned}
$$

BdG Hamiltonian
Confirming that the result actually checks out,

$$
\begin{aligned}
\left[\mathbf{H}_{\mathrm{BdG}},\boldsymbol{\rho}\right]=&\begin{bmatrix}\mathbf{H} & \boldsymbol{\Delta}\\ \boldsymbol{\Delta}^{\dagger} & -\mathbf{H}^{T}\end{bmatrix}\begin{bmatrix}\boldsymbol{\rho}_{\mathrm{ee}} & \boldsymbol{\rho}_{\mathrm{eh}}\\ \boldsymbol{\rho}_{\mathrm{he}} & \boldsymbol{\rho}_{\mathrm{hh}}\end{bmatrix}-\begin{bmatrix}\boldsymbol{\rho}_{\mathrm{ee}} & \boldsymbol{\rho}_{\mathrm{eh}}\\ \boldsymbol{\rho}_{\mathrm{he}} & \boldsymbol{\rho}_{\mathrm{hh}}\end{bmatrix}\begin{bmatrix}\mathbf{H} & \boldsymbol{\Delta}\\ \boldsymbol{\Delta}^{\dagger} & -\mathbf{H}^{T}\end{bmatrix}\\
=&\begin{bmatrix}\left(\mathbf{H}\boldsymbol{\rho}_{\mathrm{ee}}+\boldsymbol{\Delta}\boldsymbol{\rho}_{\mathrm{he}}\right) & \left(\mathbf{H}\boldsymbol{\rho}_{\mathrm{eh}}+\boldsymbol{\Delta}\boldsymbol{\rho}_{\mathrm{hh}}\right)\\ \circ & \circ\end{bmatrix}-\begin{bmatrix}\left(\boldsymbol{\rho}_{\mathrm{ee}}\mathbf{H}+\boldsymbol{\rho}_{\mathrm{eh}}\boldsymbol{\Delta}^{\dagger}\right) & \left(\boldsymbol{\rho}_{\mathrm{ee}}\boldsymbol{\Delta}-\boldsymbol{\rho}_{\mathrm{eh}}\mathbf{H}^{T}\right)\\ \circ & \circ\end{bmatrix}\\
=&\begin{bmatrix}\left(\left[\mathbf{H},\boldsymbol{\rho}_{\mathrm{ee}}\right]-\boldsymbol{\rho}_{\mathrm{eh}}\boldsymbol{\Delta}^{\dagger}+\boldsymbol{\Delta}\boldsymbol{\rho}_{\mathrm{he}}\right) & \left(\mathbf{H}\boldsymbol{\rho}_{\mathrm{eh}}+\boldsymbol{\rho}_{\mathrm{eh}}\mathbf{H}^{T}+\boldsymbol{\Delta}\boldsymbol{\rho}_{\mathrm{hh}}-\boldsymbol{\rho}_{\mathrm{ee}}\boldsymbol{\Delta}\right)\\ \circ & \circ\end{bmatrix}
\end{aligned}
$$

C. Ultra-localized orbitals approximation

Let us now consider the explicit expression of the interaction term written in an arbitrary electronic basis $\{\phi_\alpha(\mathbf{r})\}$ in real space as

$$V^{\alpha\beta}_{\gamma\delta} = \int d\mathbf{r}d\mathbf{r}'\phi^*_\alpha(\mathbf{r})\phi^*_\beta(\mathbf{r}')V(\mathbf{r}-\mathbf{r}')\phi_\gamma(\mathbf{r}')\phi_\delta(\mathbf{r}) \tag{22}$$

where we assume $V(\mathbf{r})$ to be a generic electrostatic potential with symmetry $V(\mathbf{r}-\mathbf{r}') = V(\mathbf{r}'-\mathbf{r})$. In fact, it is from this very same symmetry that the relations we initially imposed in Eqs.(**??**) come from. Furthermore, consider that we are working with a crystalline system, such that we can specify one of the generic degrees of freedom to be a position vector in the lattice $\alpha = (\mathbf{R}_\alpha, \mathbf{s}_\alpha, \tilde{\alpha})$, with $\mathbf{R}_\alpha$ specifying the position within the Bravais lattice, $\mathbf{s}_\alpha$ the respective sub-lattice and $\tilde{\alpha}$ so we don't forget other generic degrees of freedom that the system might have. Moreover, if we consider a tight-binding (TB) framework and consider the orbitals to be ultra-localized, we can reasonably assume that $V(\mathbf{r}-\mathbf{r}')$ will not vary strongly within the orbital spread and thus, we could factor out $V(\mathbf{r}-\mathbf{r}')$ from the integral in Eq.(22), allowing us to write the interaction simply as

$$V^{\alpha\beta}_{\gamma\delta} \approx V(\mathbf{r}-\mathbf{r}')\int d\mathbf{r}\phi^*_\alpha(\mathbf{r})\phi_\delta(\mathbf{r})\int d\mathbf{r}'\phi^*_\beta(\mathbf{r}')\phi_\gamma(\mathbf{r}') = V(\mathbf{r}_\alpha-\mathbf{r}_\beta)\delta_{\alpha\delta}\delta_{\beta\gamma} \tag{23}$$

where we used the orthogonal property of the orbitals and with $\mathbf{r} \equiv \mathbf{r}_\alpha$ and $\mathbf{r}' \equiv \mathbf{r}_\beta$ being the centers of the orbitals α and β respectively. Within the scope of this approximation, the self-energies defined in Eqs.(**??**)-(**??**) now simply read as

$$\Sigma^H_{ab} \approxeq V(\mathbf{r}_a-\mathbf{r}_\gamma)\rho^{\text{ee}}_{\gamma\gamma}\delta_{ab} \tag{24}$$

$$\Sigma^F_{ab} \approxeq -V(\mathbf{r}_a-\mathbf{r}_b)\rho^{\text{ee}}_{ab} \tag{25}$$

$$\Delta_{ab} \approxeq V(\mathbf{r}_a-\mathbf{r}_b)\rho^{\text{eh}}_{ab} \tag{26}$$

It is important to note that the Hartree term, contrarily to the Fock term, does not depend on the distance between the ultra-localized orbitals but rather on the purely electronic background density ρ_{ee}. One can interpret this as if the two orbitals would interact indirectly through the effects of the variations of the background density. In these sense, the Hartree term would already account for the screening dynamics of the system. This, however, cannot be said the same for the Fock term. Thus, if ones wishes to introduce some screening effect, one must add it manually; for example, by choosing, instead of a bare potential $V(\mathbf{r})$, an effective screening potential $V \to W(\mathbf{r},\omega)$. Technically, to account for the dynamics of the screening, one would have to solve the BSE self-consistently since now the Fock self-energy itself would depend on the resulting excitonic dynamics. However, as discussed in the article Ref.[Ro00], taking the static limit W($\omega = 0$) is a reasonable approximation for most semiconductor crystals since the plasmon energies that actually control the dynamic of the screening are much bigger than the excitonic binding energies. Can I say this? Does this even make sense? I'm trying to justify the use of W in my Master's Thesis. Rui, on the other hand, wrote in his notes ¨The Hartree term is only sensible to the changes in the background density and thus, due to the large range of the screened Coulomb potential, can be at first order ignored.¨ I understand the concept of taking a mean over a large sample to ignore local variations but I don't understand why I can ignore the mean all together. What's even the meaning of "first order" in the statement. Regarding the paring term, one could also specify a different potential manually, for example, thinking of it as an on-site constant Hubbard-like term $V(\mathbf{r}) \to U < 0$.